

Concentricity is not an $m = 2$ accident: the crossed-sum identity and the tent lemma at every m

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Abstract

The proof of condition (A) at $m = 2$ (2026-10-01-c2-inter-region-inequality.tex, henceforth [c2]) turned on *concentricity*: every summand of the slice sum is symmetric about one centre σ_0 independent of the summand. This paper settles whether that survives to $m \geq 3$. It does, for every m , and the mechanism is smaller than the one [c2] names.

Theorem 1.1: $\sum_i (L_i(\nu) + R_i(\nu)) = S_\nu + |\lambda|$ for every ν and every m , so on a slice every summand is symmetric about the common centre $(d - b)/2$. The two hypotheses that [c2] spends once each at $m = 2$ are the $i = 1$ and $i = 2$ instances of *one* identity, $\max(x, y + 1) + \min(x - 1, y) = x + y$, which is the horizontal-strip condition read twice; cylindric periodicity is spent exactly once more, globally, to make the single unmatched boundary term $R_m - R_0$ equal the constant n . A four-way control confirms this attribution: the two deformations the proof consumes make (C) fail, and the two it does not are silent.

Theorem 1.2 generalises the tent lemma. The support half-width is $\Lambda(\nu) = \sigma_0 - \sum_i \max(\nu_i, \lambda_{i-1} + 1)$, *separably* concave and 1-Lipschitz in each coordinate, and $\{\nu : f_\nu \neq 0\}$ is again a *box*; so the effective index set is a box-slice, exchange-connected, and the occurring half-widths form a gap-free set of consecutive integers.

What does *not* survive is the certificate. At $m = 2$ the layer profile β was the positive part of a concave function; at $m \geq 3$ it is not (24 192 of 37 448 single splines fail, and 162 aggregate slices fail, smallest $n = 9$, $\mu = (0, 1, 6)$, $\lambda = (2, 5, 8)$, $b = 2$). Worse, the route *through* β is strictly stronger than the target and is false in the natural abstract class, while the target survives there. The concavity reappears one level up: the half-width multiplicity profile $M(r) = \#\{\nu : \Lambda(\nu) = r\}$ is the positive part of a concave function on all 135 422 real slices tested. But **Theorem 5.2** shows that even this cannot close the argument: two multisets of concentric box splines with the *same* half-width profile $M = (1, 1, 1, 1)$ sum to $(1, 3, 4, 6, 4, 3, 1) \notin \text{PF}_2$ and $(1, 3, 6, 10, 6, 3, 1) \in \text{PF}_2$ respectively. So the two theorems proved here are provably *not* a complete reduction, and any proof of (A) at $m \geq 3$ must use the width vectors $w(\nu)$ and not only the half-widths they determine.

1 Setting and statements

Fix $m \geq 1$, a level $n \geq m$, and cylindric shapes $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$, stored as bead positions $\mu_1 < \dots < \mu_m < \mu_1 + n$ with $\mu_{i+m} = \mu_i + n$, likewise λ . Put $d = |\lambda/\mu| = \sum_i (\lambda_i - \mu_i)$, $\ell = 3$, and $k(a, b) = K_{\lambda/\mu, (a, b, d-a-b)}^c$. Write PF_2 for: nonnegative, support an interval of $\mathbb{Z}$, and $g(s)^2 \geq g(s-1)g(s+1)$ for all s .

By [c1, Prop. 2.3] (A-slice-sum-reformulation, proved), valid for every m : with

$$\text{Box}(\mu) = \prod_{i=1}^m [\mu_i, \mu_{i+1} - 1], \quad L_i(\nu) = \max(\nu_i, \lambda_{i-1} + 1), \quad R_i(\nu) = \min(\nu_{i+1} - 1, \lambda_i), \quad (1)$$

$$k(a, b) = \sum_{\substack{\nu \in \text{Box}(\mu) \\ S_\nu = |\mu| + b}} f_\nu(a), \quad f_\nu = \bigstar_{i=1}^m \mathbf{1}_{[L_i(\nu) - \nu_i, R_i(\nu) - \nu_i]}, \quad S_\nu = \sum_i \nu_i, \quad (2)$$

where throughout *indices are cyclic*: $\lambda_{j+m} = \lambda_j + n$ and $\nu_{j+m} = \nu_j + n$, so in particular $\lambda_0 = \lambda_m - n$ and $\nu_{m+1} = \nu_1 + n$. Values of ν with $L_i(\nu) > R_i(\nu)$ for some i contribute $f_\nu = 0$. Write

$$w_i(\nu) = R_i(\nu) - L_i(\nu) + 1, \quad \Sigma_b = \{\nu \in \text{Box}(\mu) : S_\nu = |\mu| + b\}, \quad \Sigma_b^\circ = \{\nu \in \Sigma_b : f_\nu \neq 0\}.$$

Condition (A) is the assertion that $a \mapsto k(a, b)$ is PF_2 for every b ; it is **proved** at $m = 2$ [c2, Thm. 1.1] and **computed** (open) at $m \geq 3$.

The brief for this session named one test, and it is an equivalence rather than a heuristic. A convolution $*_i \mathbf{1}_{[\alpha_i, \beta_i]}$ is symmetric about $(\sum_i \alpha_i + \sum_i \beta_i)/2$; here $\alpha_i = L_i - \nu_i$ and $\beta_i = R_i - \nu_i$, so f_ν is symmetric about

$$c(\nu) = \frac{1}{2} \left(\sum_i (L_i(\nu) + R_i(\nu)) - 2S_\nu \right), \quad (3)$$

and since S_ν is constant on a slice,

$$(C) \quad \sum_i (L_i(\nu) + R_i(\nu)) \text{ is independent of } \nu \text{ on each slice} \quad (4)$$

holds if and only if all summands on the slice share a centre.

Theorem 1.1 ((C), with its value, for every m). *For every $m \geq 1$, every cylindric $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$ and every $\nu \in \mathbb{Z}^m$,*

$$\sum_{i=1}^m (L_i(\nu) + R_i(\nu)) = S_\nu + |\lambda|.$$

Consequently every f_ν with $\nu \in \Sigma_b$ is symmetric about the common centre

$$c = \frac{1}{2} (|\lambda| - |\mu| - b) = \frac{d-b}{2},$$

and $k(\cdot, b)$ is symmetric about $(d-b)/2$. Concentricity is not an $m = 2$ accident.

Theorem 1.2 (tent lemma at every m). *Put $\sigma_0(\nu) = \frac{1}{2}(S_\nu + |\lambda|)$. For every ν with $f_\nu \neq 0$ the support of f_ν is $[c - \Lambda(\nu), c + \Lambda(\nu)]$ where*

$$\Lambda(\nu) = \sigma_0(\nu) - \sum_{i=1}^m \max(\nu_i, \lambda_{i-1} + 1) = \frac{1}{2}S_\nu - \frac{1}{2}|\lambda| + n - m - \sum_{i=1}^m (\nu_i - \lambda_{i-1} - 1)_+. \quad (5)$$

Moreover:

1. On a slice, $-\Lambda$ is a sum of m convex, nondecreasing, 1-Lipschitz functions of the single coordinates $\nu_1, \dots, \nu_m$ respectively; in particular Λ is concave on $\mathbb{R}^m$ and 1-Lipschitz in each coordinate.

2. $\Sigma_b^\circ = \Sigma_b \cap \text{Box}^\circ$ where Box° is again a box:

$$\text{Box}^\circ = \prod_{i=1}^m [\max(\mu_i, \lambda_{i-2} + 2), \min(\mu_{i+1} - 1, \lambda_i)]. \quad (6)$$

3. Hence Σ_b° is connected under the exchange moves $\nu \mapsto \nu + e_i - e_j$, along each of which $|\Delta\Lambda| \leq 1$, so $\Lambda(\Sigma_b^\circ)$ is a set of consecutive integers in the coset $\sigma_0 + \mathbb{Z}$ — it has no gaps — and Λ attains its maximum on Σ_b° .

At $m = 2$ this is [c2, Lem. 2.4] with $\Lambda = L_t$.

Theorems 1.1 and 1.2 are steps 1 and 2 of the $m = 2$ architecture, generalised. Section 4 shows that step 3 — β is the positive part of a concave function — is *false* at $m \geq 3$, and that the natural repair (prove β log-concave and quote the tail lemma) is strictly stronger than the target and false in the abstract class the two theorems above define. Section 5 isolates what remains.

2 Proof of Theorem 1.1

Everything is one scalar identity.

Lemma 2.1 (the adjacency identity). *For all $x, y \in \mathbb{Z}$ and all $p \in \mathbb{Z}$,*

$$\max(x, y + p) + \min(x - p, y) = x + y.$$

Proof. If $x \leq y$ then $x - p \leq y$ and $y + p \geq x$, so the left side is $(y + p) + (x - p) = x + y$. If $x \geq y + p$ then $x - p \geq y$, so the left side is $x + y$. If $y < x < y + p$ then $\max(x, y + p) = y + p$ and $x - p < y$ so $\min(x - p, y) = x - p$, again $x + y$. $\square$

The identity is false as soon as the two offsets differ: $\max(x, y + p) + \min(x - q, y)$ equals $x + y + p - q$ for $x \ll y$ and $x + y$ for $x \gg y$. The matched offset $p = q$ is therefore the load-bearing hypothesis, and $p = q = 1$ is exactly what (1) supplies.

Lemma 2.2 (crossed sums at every m). *For every $i \in \mathbb{Z}$ and every ν ,*

$$L_i(\nu) + R_{i-1}(\nu) = \nu_i + \lambda_{i-1},$$

where L, R are extended to all $i \in \mathbb{Z}$ by (1) and the cyclic conventions.

Proof. $L_i = \max(\nu_i, \lambda_{i-1} + 1)$ and $R_{i-1} = \min(\nu_i - 1, \lambda_{i-1})$. Apply Lemma 2.1 with $x = \nu_i$, $y = \lambda_{i-1}$, $p = q = 1$. $\square$

The two clamps paired here are $(\nu_i, \nu_i - 1)$ and $(\lambda_{i-1} + 1, \lambda_{i-1})$: each is a pair of *consecutive integers coming from one number*. That is not a coincidence of the formulas but the content of the two horizontal-strip conditions, $\nu_i \leq \kappa_i < \nu_{i+1}$ and $\kappa_{i-1} \leq \lambda_{i-1} < \kappa_i$. So Lemma 2.2 is the horizontal-strip condition read twice, once from each side, and it is bead-local: it knows nothing about m .

Proof of Theorem 1.1. Pair each L_i with R_{i-1} for $i = 1, \dots, m$. The sum $\sum_{i=1}^m R_{i-1}$ differs from $\sum_{i=1}^m R_i$ by exactly the two unmatched terms, so

$$\sum_{i=1}^m (L_i + R_i) = \sum_{i=1}^m (L_i + R_{i-1}) + (R_m - R_0) = \sum_{i=1}^m (\nu_i + \lambda_{i-1}) + (R_m - R_0) \quad (7)$$

by Lemma 2.2. Now the cyclic conventions close the cycle twice. First, $R_0(\nu) = \min(\nu_1 - 1, \lambda_0) = \min(\nu_1 + n - 1, \lambda_m) - n = R_m(\nu) - n$, so $R_m - R_0 = n$ — a constant. Second, $\sum_{i=1}^m \lambda_{i-1} = \lambda_0 + \lambda_1 + \dots + \lambda_{m-1} = |\lambda| - \lambda_m + (\lambda_m - n) = |\lambda| - n$. Substituting both into (7),

$$\sum_{i=1}^m (L_i + R_i) = S_\nu + |\lambda| - n + n = S_\nu + |\lambda|.$$

For the consequence: by (3) the centre of f_ν is $\frac{1}{2}(S_\nu + |\lambda| - 2S_\nu) = \frac{1}{2}(|\lambda| - S_\nu)$, which on Σ_b equals $\frac{1}{2}(|\lambda| - |\mu| - b) = (d - b)/2$, independent of ν . A sum of functions symmetric about a common point is symmetric about it. $\square$

2.1 Where each hypothesis is spent

The discipline of [c2] — ask *where* a hypothesis is spent, not what it implies — gives a cleaner answer here than the one [c2] itself records.

1. **The matched offset** $p = q = 1$ (equivalently: both horizontal-strip conditions) is spent once per bead, m times in all, in Lemma 2.2.
2. **Cylindric periodicity** ($\lambda_{j+m} = \lambda_j + n$ and $\nu_{j+m} = \nu_j + n$, with the *same* n) is spent exactly once, globally, on the single unmatched boundary term, to give $R_m - R_0 = n$. It is spent a second time only to evaluate the constant, $\sum_i \lambda_{i-1} = |\lambda| - n$.

[c2] describes the $m = 2$ proof as spending two hypotheses, $A = D - n + 1$ and $C = B + 1$, once each. That is the $m = 2$ shadow of item 1: in the dictionary $a_t = L_1$, $b_t = R_1$, $c_t = L_2$, $d_t = R_2$ (with $t = \nu_1$, $u = S_\nu$), the identity $b_t + c_t = u + B - t$ is the $i = 2$ instance of Lemma 2.2 and $a_t + d_t = t + D$ is the $i = 1$ instance together with the wrap $R_0 = R_m - n$. So there is one local identity used m times, plus one global closure — and at $m = 2$ the global closure happens to be invisible, because with only two beads it hides inside the second crossed sum.

Remark 2.3 (a correction to my own first reading). On first deriving Lemma 2.2 I concluded that periodicity is *not* load-bearing for (C), on the ground that λ_0 enters only the additive constant. That is wrong, and the control in §6 said so. The error was to look at $\sum_i \lambda_{i-1}$ and forget $R_m - R_0$: periodicity is what makes the one unmatched pair differ by a constant. Decoupling the λ -period from the ν -period breaks (C) (control P3 below), while changing the offset *value* $p = q$ does not (control P1).

3 Proof of Theorem 1.2

Proof. The formula (5). If $f_\nu \neq 0$ then every $w_i \geq 1$ and f_ν is supported on an interval of length $\sum_i w_i - m + 1$, so its half-width is $\Lambda = \frac{1}{2}(\sum_i w_i - m)$. By Theorem 1.1, $\sum_i R_i = S_\nu + |\lambda| - \sum_i L_i$, hence $\sum_i w_i = \sum_i R_i - \sum_i L_i + m = S_\nu + |\lambda| - 2\sum_i L_i + m$ and

$$\Lambda(\nu) = \frac{1}{2}(S_\nu + |\lambda|) - \sum_i L_i(\nu) = \sigma_0(\nu) - \sum_i \max(\nu_i, \lambda_{i-1} + 1),$$

which is the first form. For the second, write $\max(\nu_i, \lambda_{i-1} + 1) = (\lambda_{i-1} + 1) + (\nu_i - \lambda_{i-1} - 1)_+$ and use $\sum_i (\lambda_{i-1} + 1) = |\lambda| - n + m$.

(1). $\phi_i(x) := \max(x, \lambda_{i-1} + 1)$ is convex, nondecreasing and 1-Lipschitz on $\mathbb{Z}$, with increments $\phi_i(x+1) - \phi_i(x) \in \{0, 1\}$, equal to 1 exactly when $x \geq \lambda_{i-1} + 1$. On a slice σ_0 is a constant, so $\Lambda = \text{const} - \sum_i \phi_i(\nu_i)$ is a constant minus a separable convex function.

(2). $f_\nu \neq 0$ iff $L_i(\nu) \leq R_i(\nu)$ for every i , i.e. iff $\max(\nu_i, \lambda_{i-1} + 1) \leq \min(\nu_{i+1} - 1, \lambda_i)$, which is the conjunction of four inequalities:

$$\nu_i \leq \nu_{i+1} - 1, \quad \nu_i \leq \lambda_i, \quad \lambda_{i-1} + 1 \leq \nu_{i+1} - 1, \quad \lambda_{i-1} + 1 \leq \lambda_i.$$

The first holds automatically on $\text{Box}(\mu)$, since $\nu_i \leq \mu_{i+1} - 1 \leq \nu_{i+1} - 1$; the fourth holds because λ is a shape. The second and third are each a bound on a *single* coordinate; reindexing the third ($i+1 \rightsquigarrow i$) it reads $\nu_i \geq \lambda_{i-2} + 2$. Intersecting with $\text{Box}(\mu)$ gives (6).

(3). Σ_b° is the set of integer points of a box lying on the hyperplane $\sum_i x_i = |\mu| + b$. Such a set is connected under the moves $x \mapsto x + e_i - e_j$: if $x \neq y$ lie in it, then $\sum_i (x_i - y_i) = 0$, so

there are i with $x_i < y_i$ and j with $x_j > y_j$, and $x + e_i - e_j$ is again in the box (each coordinate moves one step towards y_i resp. y_j , hence stays between the box bounds) and on the hyperplane, strictly reducing $\sum_i |x_i - y_i|$. Along such a move

$$\Delta\Lambda = -[\phi_i(x_i + 1) - \phi_i(x_i)] + [\phi_j(x_j) - \phi_j(x_j - 1)] \in \{-1, 0, 1\}$$

because each bracket lies in $\{0, 1\}$. A function with increments in $\{-1, 0, 1\}$ along the edges of a connected graph has an image with no gaps, and the image is finite, so the maximum is attained. All values lie in $\sigma_0 + \mathbb{Z}$ since each ϕ_i is $\mathbb{Z}$ -valued. $\square$

Remark 3.1. Part (2) is the general- m replacement for the $m = 2$ interval $T = [\max(\mu_1, u - \mu_1 - n + 1), \min(\mu_2 - 1, u - \mu_2)]$ of [c2, eq. (3)]: at $m = 2$ the effective index set is a box-slice in two coordinates, i.e. an interval in $t = \nu_1$, and that is where [c2] got its $T_{\pm}$. Part (2) explains why no extra “ $h_t \leq 0$ ” case analysis is needed: the degenerate ν are cut off by *box* constraints, not by a coupled condition.

4 The $m = 2$ certificate is dead at $m \geq 3$

By Theorem 1.1, on a slice we may write $x = a - c$ and

$$G(a) := k(a, b) = \gamma(|x|), \quad \gamma(y) = G(c + y),$$

with γ nonnegative and nonincreasing (each f_{ν} is a convolution of interval indicators, hence symmetric and unimodal about c). Define the *layer profile*

$$\beta(r) := \gamma(r) - \gamma(r + 1) \geq 0, \quad \text{so} \quad f = \sum_r \beta(r) \mathbf{1}_{[c-r, c+r]}, \quad \gamma(y) = \sum_{r \geq y} \beta(r).$$

Each individual f_{ν} has its own layer profile β_{ν} , and $\beta = \sum_{\nu \in \Sigma_b^{\circ}} \beta_{\nu}$. By the tail lemma [c2, Lem. 4.1] (**tail-of-pf2-logconcave, proved**): *if β is log-concave with interval support then γ is log-concave*, and then by the radial lemma [c2, Lem. 4.3] G is PF₂. So condition (A) at general m follows from

$$(B) \quad \beta \text{ is log-concave with interval support.}$$

At $m = 2$, [c2, Prop. 3.2] proves the much stronger statement that $\beta = c_+$ for an explicit concave c . That is what dies.

Proposition 4.1 (step 3 of [c2] fails at $m \geq 3$). *1. For a single width vector $w \in \mathbb{Z}_{\geq 1}^m$, the layer profile β_w of $*_i \mathbf{1}_{[0, w_i - 1]}$ is log-concave in all 37 448 cases with $m \leq 5$, $w_i \leq 8$, but is not the positive part of a concave function in 24 192 of them. The smallest witness is $w = (3, 3, 3, 3)$, where $\beta_w = (3, 6, 6, 3, 1)$ (innermost layer first): log-concave, with increments $+3, 0, -3, -2$, hence not concave.*

2. For the aggregate β on a real slice, $\beta = c_+$ fails on 162 of the 44 173 nonzero slices with $m = 3$, $n \leq 10$, $d \leq 12$. The smallest by d is $n = 9$, $\mu = (0, 1, 6)$, $\lambda = (2, 5, 8)$, $b = 2$, where Σ_b° has three elements:

ν	$(L_i, R_i)_i$	$w(\nu)$	f_{ν}	β_{ν}
$(0, 1, 8)$	$(0, 0), (3, 5), (8, 8)$	$(1, 3, 1)$	$(1, 1, 1)$ on $[2, 4]$	$(0, 1)$
$(0, 2, 7)$	$(0, 1), (3, 5), (7, 8)$	$(2, 3, 2)$	$(1, 3, 4, 3, 1)$ on $[1, 5]$	$(1, 2, 1)$
$(0, 3, 6)$	$(0, 2), (3, 5), (6, 8)$	$(3, 3, 3)$	$(1, 3, 6, 7, 6, 3, 1)$ on $[0, 6]$	$(1, 3, 2, 1)$

all three concentric about $c = 3$ (consistently with Theorem 1.1: $d = 8$, $b = 2$), with $G = (1, 4, 10, 12, 10, 4, 1)$ and $\beta = (2, 6, 3, 1)$. Each β_ν is concave; the sum is not, because the positive-part-of-concave class is not closed under addition when the supports differ. G is nevertheless PF_2 and β is log-concave.

3. The pairwise route fails at the layer level too. On the instance of (2), $\beta_{(1,3,1)} = (0, 1, 0, 0)$ and $\beta_{(3,3,3)} = (1, 3, 2, 1)$ violate $2g(r)h(r) \geq g(r-1)h(r+1) + g(r+1)h(r-1)$ at $r = 2$: $0 \geq 1$. Over $m \leq 5$, $n \leq 9$, $d \leq 11$ there are 2867 failing layer pairs out of 84051. So the antichain obstruction of [c1, §2.1 item 1] is not removed by passing to layers.
4. β is not real-rooted (e.g. $\beta = (1, 1, 1)$ at $n = 6$, $\mu = (0, 1)$, $\lambda = (2, 5)$, $b = 2$), so no interlacing argument is available — the same obstruction as [c1, §2.1 item 3], one level down.

Proof. All four are finite computations; see §7. The structural remarks are checked by hand on the displayed data: for (1), $(1, 1, 1)^{*4} = (1, 4, 10, 16, 19, 16, 10, 4, 1)$ has $\gamma = (19, 16, 10, 4, 1)$ and $\beta = (3, 6, 6, 3, 1)$; for (2), $\gamma = (12, 10, 4, 1)$ gives $\beta = (2, 6, 3, 1)$ with increments $+4, -3, -2$, and $(0, 1) + (1, 2, 1) + (1, 3, 2, 1) = (2, 6, 3, 1)$. $\square$

5 The gap, stated so that it has no quantifier left to shrink

Theorems 1.1 and 1.2 say that the slice family is a family of *concentric box splines whose half-widths are gap-free*. Does that abstract datum suffice? No — and the obstruction is not that the datum is too weak by a little. It is that the datum is *shape-blind*, and no shape-blind hypothesis can work.

Proposition 5.1 (three successively stronger abstract hypotheses, all insufficient). *Let $g_1, \dots, g_N$ be convolutions of interval indicators, all centred at 0, with half-widths $\Lambda_1, \dots, \Lambda_N$ and multiplicity profile $M(r) = \#\{j : \Lambda_j = r\}$. Each g_j is PF_2 .*

1. **Concentricity alone is insufficient.** $g_1 = \delta_0$ ($w = (1)$) and $g_2 = \mathbf{1}_{[-2,2]}$ ($w = (5)$) give $g_1 + g_2 = (1, 1, 2, 1, 1)$, not PF_2 ; here $M = (1, 0, 1)$ has a gap.
2. **Adding gap-freeness is insufficient.** Three copies of δ_0 , one $\mathbf{1}_{[-1,1]}$ and one $\mathbf{1}_{[-2,2]}$ give $(1, 2, 5, 2, 1)$, not PF_2 , with $M = (3, 1, 1)$: gap-free, but not log-concave, since $1^2 < 3 \cdot 1$.
3. **Adding “ M is the positive part of a concave function” is insufficient.** At $m = 2$ take $w^{(1)} = (1, 1)$, $w^{(2)} = (2, 2)$, $w^{(3)} = (1, 5)$, $w^{(4)} = (2, 6)$, i.e.

$$\delta_0, \quad (1, 2, 1), \quad (1, 1, 1, 1, 1), \quad (1, 2, 2, 2, 2, 2, 1),$$

with half-widths 0, 1, 2, 3, so $M = (1, 1, 1, 1)$ — flat, hence concave. The sum is $(1, 3, 4, 6, 4, 3, 1)$, and $4^2 = 16 < 3 \cdot 6 = 18$: not PF_2 .

Proof. Each item is the displayed arithmetic. (1) $\delta_0 + \mathbf{1}_{[-2,2]} = (1, 1, 2, 1, 1)$ and $1^2 < 1 \cdot 2$ at the points adjacent to the centre. (2) $3\delta_0 + \mathbf{1}_{[-1,1]} + \mathbf{1}_{[-2,2]} = (1, 2, 5, 2, 1)$ and $2^2 < 1 \cdot 5$. (3) summing the four displayed sequences after centring gives $(1, 3, 4, 6, 4, 3, 1)$. $\square$

Item (3) is not merely one more counterexample; it closes the whole family.

Theorem 5.2 (no shape-blind hypothesis can suffice). *There exist two multisets of concentric convolutions of interval indicators with the same half-width multiplicity profile $M = (1, 1, 1, 1)$ — the flattest and most concave profile available — such that one sum is PF_2 and the other is not:*

$$\begin{aligned} \{(1, 1), (2, 2), (1, 5), (2, 6)\} &\longrightarrow (1, 3, 4, 6, 4, 3, 1) \text{ not } \text{PF}_2, \\ \{(1, 1), (2, 2), (3, 3), (4, 4)\} &\longrightarrow (1, 3, 6, 10, 6, 3, 1) \text{ } \text{PF}_2. \end{aligned}$$

Consequently no condition formulated in terms of the multiset $\{\Lambda_j\}$ alone — no matter how strong — implies $\sum_j g_j \in \text{PF}_2$. Any proof of condition (A) at $m \geq 3$ must use the width vectors $w(\nu)$ themselves, not only the half-widths $\Lambda(\nu) = \frac{1}{2}(\sum_i w_i(\nu) - m)$ they determine.

Proof. Both multisets have half-widths 0, 1, 2, 3, hence the same M . The first sums to $\delta_0 + (1, 2, 1) + (1, 1, 1, 1, 1) + (1, 2, 2, 2, 2, 1) = (1, 3, 4, 6, 4, 3, 1)$, which fails PF_2 at $16 < 18$. The second sums to $\delta_0 + (1, 2, 1) + (1, 2, 3, 2, 1) + (1, 2, 3, 4, 3, 2, 1) = (1, 3, 6, 10, 6, 3, 1)$, which satisfies $6^2 = 36 \geq 3 \cdot 10 = 30$ and $10^2 \geq 6^2$, and has interval support. A hypothesis that is a function of M alone takes the same value on both, so it either admits the first (and is not sufficient) or excludes the second (and is not satisfied by all real slices, since $M = (1, 1, 1, 1)$ does occur). $\square$

Remark 5.3 (what the width vectors supply at $m = 2$, and the shape of the open question). Theorem 5.2 explains why the $m = 2$ proof could not be shape-blind and was not. There the width pair obeys $w_1(\nu) - w_2(\nu) = 2(t_0 - \nu_1)$ ([c2, eq. (6)]), so the *difference* determines the slice parameter: no two summands of one slice can share it. The first multiset of Theorem 5.2 has differences 0, 0, -4 , -4 and is therefore not realisable at $m = 2$ — the coupling between shape and position is exactly what rules it out. The open question at $m \geq 3$ is thus not “which condition on Λ ” but:

which m -tuples $w(\nu)$ can co-occur in one slice, and why does that family sum to a PF_2 sequence?

Each $w_i(\nu) = \min(\nu_{i+1} - 1, \lambda_i) - \max(\nu_i, \lambda_{i-1} + 1) + 1$ is concave in each of the two coordinates it sees, and $\sum_i w_i$ is pinned by Theorem 1.1; what is missing is a usable description of the joint law of the w_i along a slice.

Remark 5.4 (the route through (B) is strictly too strong). It is tempting to prove (B) — β log-concave — and quote the tail lemma, and (B) does hold on every real slice tested (§7): 4 323 908 slices, including $m = 3$ with $n \leq 16$, $d \leq 20$. But (B) is *false* in the abstract class of Proposition 5.1(2) while the target is *true* there: for $N = 2$, $m = 2$, $w^{(1)} = (1, 5)$ and $w^{(2)} = (3, 3)$ (half-widths 2, 2, gap-free), the sum is $(1, 1, 1, 1, 1) + (1, 2, 3, 2, 1) = (2, 3, 4, 3, 2)$, which *is* PF_2 , while $\beta = (1, 1, 2)$ is *not* log-concave, since $1^2 < 1 \cdot 2$. In the exhaustive abstract search of §7, (B) fails on 23 gap-free multisets at $m = 2$ and 198 at $m = 3$, and the PF_2 conclusion fails on *none* of those. So (B) cannot be derived from Theorems 1.1 and 1.2: it is a true statement about real slices that is strictly stronger than what those two theorems carve out, and proving it requires the same missing input as proving (A) directly, plus more. For a proof it is therefore the wrong intermediate; for a *test* it is a good one, because it is sharper than (A) and so falsifies faster.

Remark 5.5 (where the concavity went). One positive structural fact survives and is worth recording even though Theorem 5.2 shows it cannot be sufficient on its own: on real slices the half-width multiplicity profile M *is* the positive part of a concave function, on all 135 422 nonzero slices with $2 \leq m \leq 5$, $m \leq n \leq m + 5$, $d \leq 11$. So the concavity that the $m = 2$ proof spends at the *layer* level — where it is false at $m \geq 3$, Proposition 4.1 — reappears one level up, at the

half-width level, where it is a consequence to be proved from Theorem 1.2 rather than an input. By Theorem 5.2 it will have to be combined with shape information, but it is a genuine invariant of the family and the first thing I would try to prove next.

6 The control

The brief asked for a refusal control: break the cylindric alignment and confirm that (C) fails. The proof of Theorem 1.1 says more than that — it says *which* deformations must break (C) and which must not — so the control is run as a four-way panel with all four outcomes predicted in advance from (7). Deform (1) to

$$L_i^{(p)}(\nu) = \max(\nu_i, \lambda_{i-1} + p), \quad R_i^{(q)}(\nu) = \min(\nu_{i+1} - q, \lambda_i),$$

and let the λ -period be n' while the ν -period stays n .

	deformation	prediction	$m = 3$ outcome
P1	$p = q = 2$ and $p = q = 4$ (offset <i>value</i>)	silent	(C) holds
P2	$p = 2, q = 1$ and $p = 1, q = 3$ (offset <i>mismatch</i>)	fires	fails 792/1306, 765/1306
P3	$n' = n \pm 1, n' = n + 2$ (periods decoupled)	fires	fails 295/1306, 396, 377
P4	no wrap: $L_1 = \nu_1, R_m = \lambda_m$ (ordinary skew)	silent	(C) holds

All four predictions hold, at $m = 2, 3, 4$ (counts in §7). P1 is silent because Lemma 2.1 holds for every matched offset; P2 fires because the local collapse is destroyed; P3 fires because the unmatched boundary term $R_m - R_0$ becomes ν -dependent; P4 is silent because in the linear case the two boundary terms degenerate *separately* — $L_1 = \nu_1$ is absorbed into S_ν and $R_m = \lambda_m$ is a constant — which is the statement that ordinary skew Kostka numbers also have concentric slice sums.

A control that fires on everything would not have distinguished the offset match from the periodicity, and it was P3 that corrected my first (wrong) attribution; a control that fired on nothing would have said the hypotheses are not load-bearing. This one fires on exactly the two deformations the proof consumes.

7 Verification

Code in `proofs/code-1002-m3/`. No instrument's output was believed before it reproduced a value already known.

Calibration.

- The general- m slice-sum engine reproduces both known $m = 2$ values exactly, with offsets: $k(\cdot, 2) = (1, 3, 6, 7, 6, 3, 1)$ at $n = 8, \mu = (0, 3), \lambda = (4, 7)$ ([c1, §“Concavity is false”]) and $G = (1, 4, 7, 10, 11, 10, 7, 4, 1)$ at $n = 10, \mu = (0, 5), \lambda = (5, 12), b = 4$ ([c2, Prop. 2.4]).
- **Estimand check.** An independent enumerator builds the chains $\mu \prec \nu \prec \kappa \prec \lambda$ directly from `is_shape`, `hstrip`, `size` of `proofs/code-q254/cyl.py`, never touching L_i, R_i . It agrees with (2) on all 13 689 slices tested: $m = 2$ ($n \leq 7, d \leq 9$; 2415), $m = 3$ ($n \leq 7, d \leq 8$; 5124), $m = 4$ ($n \leq 7, d \leq 7$; 4150), zero disagreements. The same independent table confirms $k(a, b) = k(b, a)$ and $k(a, b) = k(d - a - b, b)$ on 14 833 entries, zero bad — the latter being the total-sum shadow of Theorem 1.1.

- **Refusal probes.** The PF_2 test refuses $(1, 1, 2, 1, 1)$; the concave-positive-part test refuses $(3, 6, 6, 3, 1)$; the four perturbed forms of (5) and (6) ($\lambda_{i-1} + 1 \mapsto \lambda_{i-1} + 0, +2$; $\lambda_{i-2} + 2 \mapsto \lambda_{i-2} + 1, +3$) are refused on 229/466, 466/466, 400/1047, 434/1047 instances respectively.

Results.

- **Theorem 1.1, STEP 0 census.** $\sum_i (L_i + R_i)$ is constant on every one of 8335376 nonempty slices, and equals $S_\nu + |\lambda|$ on every one of 40735322 summands — the closed form, not merely constancy — over $m = 2$ ($n \leq 16, d \leq 20$), $m = 3$ ($n \leq 14, d \leq 18$), $m = 4$ ($n \leq 14, d \leq 18$), $m = 5$ ($n \leq 12, d \leq 14$), $m = 6$ ($n \leq 11, d \leq 12$). The range is deliberately wider than the $m = 2$ range of $[c2]$ in every parameter.
- **Theorem 1.2.** (5) is exact on all 43916 nonzero f_ν and (6) is exact on all 94906 ν tested ($2 \leq m \leq 5, n \leq m + 4, d \leq 8$). Gap-freeness of $\Lambda(\Sigma_b^\circ)$: 78312 of 78312 slices ($2 \leq m \leq 5, n \leq 9, d \leq 11$).
- **Proposition 4.1.** (1) 37448 width vectors, $m \leq 5, w_i \leq 8$: β_w log-concave in all, $= c_+$ in 13256 and not in 24192. (2) 44173 slices at $m = 3, n \leq 10, d \leq 12$: $\beta = c_+$ fails on 162. (3) 2867 failing layer pairs of 84051. (4) β real-rooted on 74669 of 78312.
- **Condition (A) and (B), range-extended.** (A) and (B) both hold with zero failures on every slice tested: $m = 3, n \leq 16, d \leq 20$ (1519862 slices); $m = 4, n \leq 13, d \leq 16$ (1502376); $m = 5, n \leq 12, d \leq 13$ (1069479); $m = 6, n \leq 11, d \leq 12$ (232191); $m = 7$ in progress, see `code-1002-m3/wide.out` for the running counts. ($\beta = c_+$ for a concave c fails on 105084 of the $m = 3$ slices, 57877 of the $m = 4$, 14870 of the $m = 5$ and 60 of the $m = 6$ — Proposition 4.1(2) at scale.) This extends the registry’s 96781 interior points ($n \leq 9, m \leq 7, d \leq 11$) by more than an order of magnitude in $m = 3, 4$. G was symmetric and $\beta \geq 0$ on every slice, as Theorem 1.1 requires.
- **Proposition 5.1 and Remark 5.4.** Exhaustive over multisets of concentric box splines: at $m = 1$ with $w \leq 9$ and size ≤ 6 , gap-freeness gives zero PF_2 failures at sizes 2, 3, 4 and **13 failures at size 5, 32 at size 6**; at $m = 2$ with $w_i \leq 5$, zero at sizes 2, 3 and 5 failures at size 4, 21 at size 5. *The discovery set was sizes ≤ 3 , where the score was perfect.* The (B) failures in the gap-free class are 23 at $m = 2$ and 198 at $m = 3$, with the PF_2 conclusion failing on none of those.
- **Theorem 5.2 and Proposition 5.1(3).** Restricting the abstract search to multisets whose M is the positive part of a concave function: $m = 1, w \leq 11$, sizes 2–8 (20, 36, 50, 71, 104, 127, 167 multisets) gives **zero** PF_2 failures; $m = 2, w_i \leq 6$, sizes 2, 3 (216, 1238) gives zero, and size 4 (4606) gives **4 failures**, all equivalent to Proposition 5.1(3) under the symmetry $w \mapsto \text{reversal}$; $m = 3, w_i \leq 5$, sizes 2, 3 (2444, 45369) gives zero. Again the discovery set — sizes ≤ 3 — scores perfectly and the statement is false at size 4. The two multisets of Theorem 5.2 were then checked directly.
- **Proposition 5.1(3).** $M = c_+$ for a concave c on all 135422 nonzero slices with $2 \leq m \leq 5, m \leq n \leq m + 5, d \leq 11$; M fails to be nondecreasing on 10000 of them and fails to be nonincreasing on 11966, so the concavity is not an artefact of monotonicity.

8 Gaps, and what comes next

1. **No gap in Theorem 1.1 or Theorem 1.2.** Both are proved from Lemma 2.1, the cyclic conventions, and the connectivity of a box-slice. The one external input is [c1, Prop. 2.3], proved.
2. **Condition (A) at $m \geq 3$ is still open, and is now known not to be shape-blind.** Theorem 5.2 rules out every hypothesis phrased in the half-widths alone, so Theorems 1.1 and 1.2 are provably not a complete reduction. Two concrete next targets, in order: (i) prove that M is the positive part of a concave function (Remark 5.5) — $M(r)$ counts the lattice points of a box-slice on which the separably convex $\sum_i \phi_i(\nu_i)$ takes a fixed value, and Λ is 1-Lipschitz along exchanges by Theorem 1.2(3), so this should be accessible; (ii) describe which width vectors $w(\nu)$ co-occur on a slice, which is the input Theorem 5.2 says is unavoidable.
3. **Three steps of the $m = 2$ proof, at $m \geq 3$.** Step 1 (concentricity): **proved**, Theorem 1.1. Step 2 (the tent): **proved**, Theorem 1.2. Step 3 ($\beta = c_+$ for concave c): **false**, Proposition 4.1, and the concavity relocates to M — where, by Theorem 5.2, it is still not sufficient. Step 4 (tail lemma + radial lemma): **unchanged**, both are statements about one sequence and are already proved for all m ; they apply the moment (B) is available, but Remark 5.4 shows (B) is the wrong thing to aim at.
4. **Condition (B) (conj-B-logsubmodular)** is untouched and remains computed.
5. **A by-product worth recording.** Control P4 shows that concentricity holds for *ordinary* skew shapes too, by a degeneration of the same telescoping. So the mechanism is not cylindric; what is cylindric is only the form of the boundary term.

9 Summary

The brief asked whether concentricity is an $m = 2$ accident or a cylindric fact. It is neither: it is a *horizontal-strip* fact. One scalar identity, $\max(x, y + 1) + \min(x - 1, y) = x + y$, applied once per bead, plus one global closure of the cycle, gives it at every m , and a four-way control confirms that attribution by firing on exactly the two deformations the proof consumes. The tent lemma generalises too, and more cleanly than at $m = 2$: the half-width is a separably concave function on a box-slice, so it is gap-free for the standard exchange-connectivity reason.

What does not generalise is the certificate, and the session's sharpest finding is negative. The layer profile is no longer the positive part of a concave function; the route through its log-concavity is strictly stronger than the target and false in the abstract class the two theorems carve out; and the concavity reappears one level up, at the half-width multiplicity profile — where Theorem 5.2 shows it *still* cannot suffice, because two multisets with the same half-width profile can land on opposite sides of PF_2 . That last result is the one I would keep: it says the two theorems proved here are provably not a complete reduction, and that any proof of condition (A) at $m \geq 3$ must know which width vectors co-occur in a slice. I had intended to leave a conjecture in its place; the conjecture was refuted by the search I launched to test it, at multiset size 4 after sizes 2 and 3 scored perfectly. That is the third session in a row in which extending a range past the one that bore the statement was the decisive move, and the first in which it caught a conjecture of mine before I wrote it down as one.